THE GEOMETRIC GENUS OF A PROPER CURVE IS BOUNDED BY THE SEPARABLE DEGREE TIMES THE FIRST COHOMOLOGY DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the separable-degree fibre decomposition, the flat base change matching, the normalization decomposition, and the genus-goes-down summation are all retained verbatim.

Editorial note. The label lemma-bound-geometric-genus-curve names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Global functions and separable degree. Varieties, Lemma varieties-lemma-regular-functions-proper-variety; Fields, Lemma fields-lemma-separable-degree. Makes kappa = H^0(C, O_C) a finite extension of k and identifies [kappa : k]_s with the number of k-embeddings into the algebraic closure, i.e. the number of geometric fibres.
- Base change and normalization decomposition. Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology; Morphisms, Lemma morphisms-lemma-normalization-in-terms-of-components. Preserves dim H^1 under the field base change and decomposes the normalization of each geometric fibre into nonsingular projective models of its components.
- Genus goes down under normalization. Same chapter, Lemma lemma-genus-goes-down. Supplies the final inequality dim H^1(C_t, O) >= dim H^1(C_t^nu, O) comparing the curve with its normalization.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Prerequisite pointer invoked at proof line(s) 3943; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09HJ: sources/stacks-chow/tags.txt, source lines 12007–12007. Prerequisite pointer invoked at proof line(s) 3921; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CE3: sources/stacks-curves/pinned-original-source.tex, source lines 3850–3850. Prerequisite pointer invoked at proof line(s) 3950; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0CDV: sources/stacks-chow/tags.txt, source lines 15555–15555. Prerequisite pointer invoked at proof line(s) 3947; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04L2: sources/stacks-chow/tags.txt, source lines 5993–5993. Prerequisite pointer invoked at proof line(s) 3919; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let C be a proper curve over k. Set $\kappa = H^0(C, \mathcal{O}_C)$. Then*

$$[\kappa : k]_s \dim_\kappa H^1(C, \mathcal{O}_C) \geq g_{geom}(C/k)$$

Proof. Varieties, Lemma 04L2 implies κ is a field and a finite extension of k. By Fields, Lemma 09HJ we have $[\kappa : k]_s = |\operatorname{Mor}_k(\kappa, \overline{k})|$ and hence $\operatorname{Spec}(\kappa \otimes_k \overline{k})$ has $[\kappa : k]_s$ points each with residue field $\overline{k}$. Thus

$$C_{\overline{k}} = \bigcup\nolimits_{t \in \operatorname{Spec}(\kappa \otimes_k \overline{k})} C_t$$

(set theoretic union). Here $C_t = C \times_{\operatorname{Spec}(\kappa), t} \operatorname{Spec}(\overline{k})$ where we view t as a k-algebra map $t : \kappa \to \overline{k}$. The conclusion is that $g_{geom}(C/k) = \sum_t g_{geom}(C_t/\overline{k})$ and the sum is over an index set of size $[\kappa : k]_s$. We have

$$H^0(C_t, \mathcal{O}_{C_t}) = \overline{k} \quad \text{and} \quad \dim_{\overline{k}} H^1(C_t, \mathcal{O}_{C_t}) = \dim_\kappa H^1(C, \mathcal{O}_C)$$

by cohomology and base change (Cohomology of Schemes, Lemma 02KH). Observe that the normalization C_t^ν is the disjoint union of the nonsingular projective models of the irreducible components of C_t (Morphisms, Lemma 0CDV). Hence $\dim_{\overline{k}} H^1(C_t^\nu, \mathcal{O}_{C_t^\nu})$ is equal to $g_{geom}(C_t/\overline{k})$. By Lemma 0CE3 we have

$$\dim_{\overline{k}} H^1(C_t, \mathcal{O}_{C_t}) \geq \dim_{\overline{k}} H^1(C_t^\nu, \mathcal{O}_{C_t^\nu})$$

and this finishes the proof. □